

Deviated Pursuit Guidance Analysis: Parameter Variation Studies

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Nov 19 2025

1 Experimental Design: Parameter Variation Groups

The simulation scenarios are strategically organized into three distinct parameter variation groups, plus two critical low-speed-ratio cases for boundary analysis.

Table 1: Parameter Variation Experimental Design

Group	Scenarios	Variable Parameter	Constant Parameters	Values
I	1-3	Deviation Angle (δ)	$V_M = 800$, $V_T = 450$ $\theta = 30^\circ$, $\alpha_T = 45^\circ$	5° , 15° , 25°
II	4-6	Initial LOS (θ)	$V_M = 700$, $V_T = 500$ $\delta = 10^\circ$, $\alpha_T = 45^\circ$	15° , 30° , 45°
III	7-9	Target Flight Path (α_T)	$V_M = 600$, $V_T = 450$ $\theta = 30^\circ$, $\delta = 10^\circ$	30° , 45° , 60°
IV	10-11	Low Speed Ratio	Boundary analysis	Critical limits

2 Group I Analysis: Deviation Angle Variation (δ)

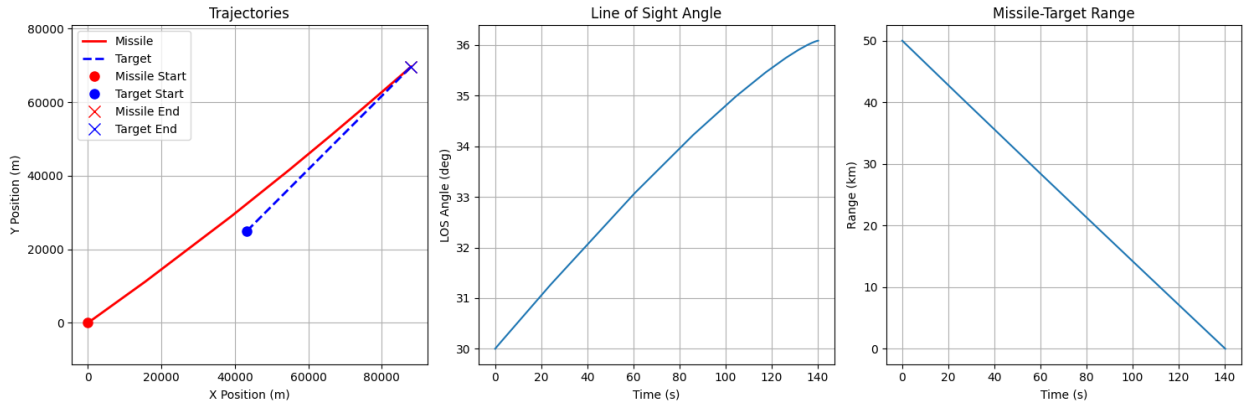
2.1 Experimental Conditions

- Constant parameters: $V_M = 800 \text{ m s}^{-1}$, $V_T = 450 \text{ m s}^{-1}$, $K = 1.778$
- Initial conditions: $\theta = 30^\circ$, $\alpha_T = 45^\circ$, $R_{init} = 50 \text{ km}$
- Variable: $\delta = 5^\circ, 15^\circ, 25^\circ$

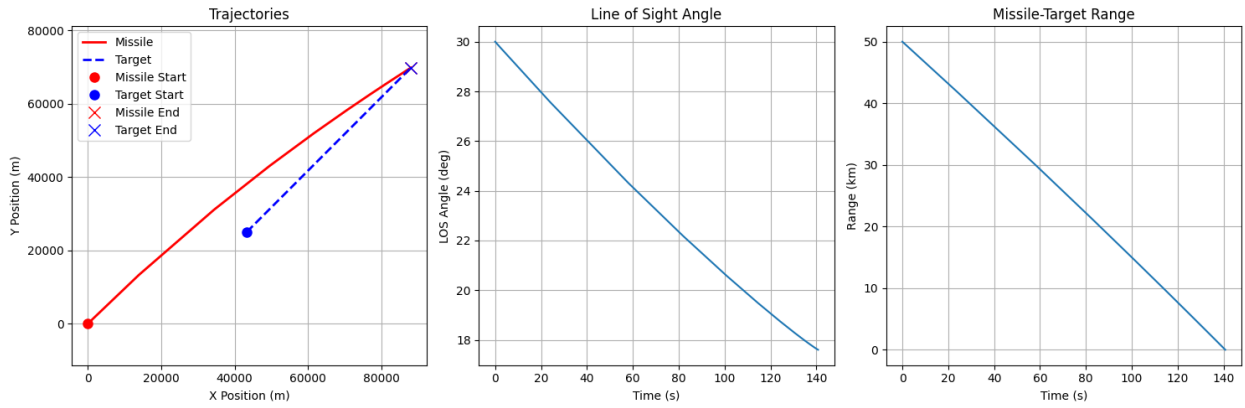
Table 2: Group I: Deviation Angle Impact Results

Scenario	δ ($^\circ$)	Intercept Time (s)	Final LOS ($^\circ$)	Efficiency
3	5	140.29	36.09	Excellent
1	15	140.76	17.60	Excellent
2	25	144.35	-3.70	Very Good

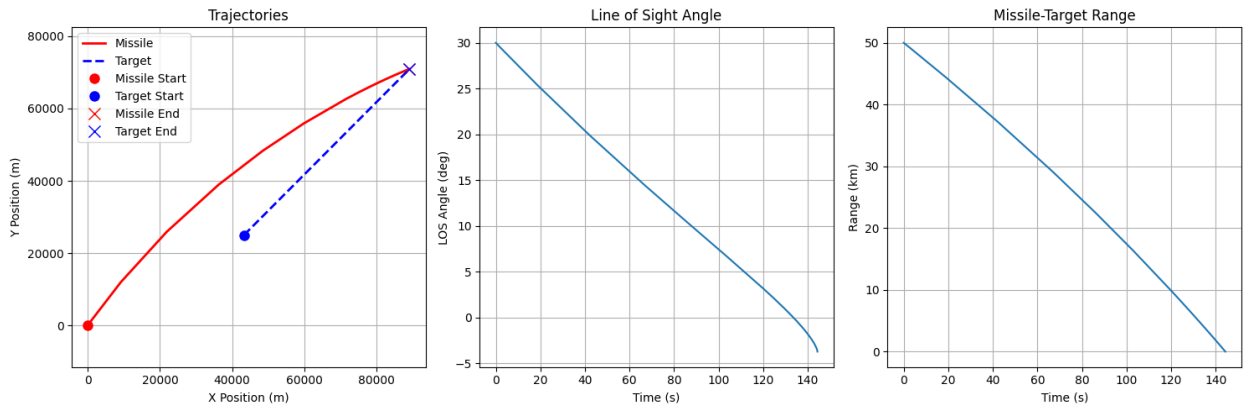
2.2 Key Findings - Group I



(a) $\delta = 5^\circ$ (Scenario 3)



(b) $\delta = 15^\circ$ (Scenario 1)



(c) $\delta = 25^\circ$ (Scenario 2)

Figure 1: Trajectory comparison for different deviation angles (Group I)

Deviation Angle Effects:

- **Intercept Time:** Minimal variation (140-144 seconds) due to high speed ratio dominance
- **Final LOS Angle:** Strong correlation with deviation angle

- $\delta = 5^\circ$: Final LOS = 36.09° (overshoot)
- $\delta = 15^\circ$: Final LOS = 17.60° (balanced)
- $\delta = 25^\circ$: Final LOS = -3.70° (undershoot then correction)
- **Trajectory Curvature:** Higher δ creates more curved paths requiring more lateral acceleration

3 Group II Analysis: Initial LOS Angle Variation (θ)

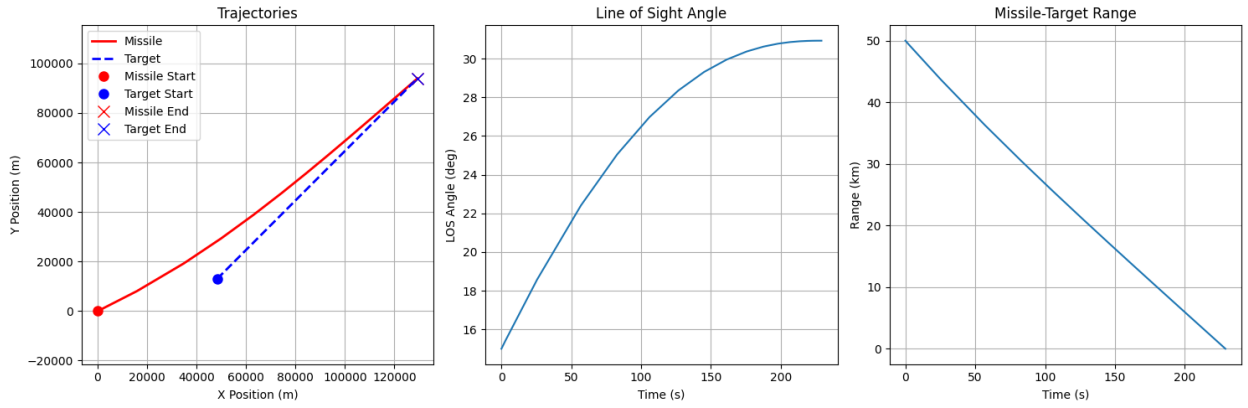
3.1 Experimental Conditions

- Constant parameters: $V_M = 700 \text{ m s}^{-1}$, $V_T = 500 \text{ m s}^{-1}$, $K = 1.400$
- Guidance parameters: $\delta = 10^\circ$, $\alpha_T = 45^\circ$
- Variable: $\theta = 15^\circ, 30^\circ, 45^\circ$

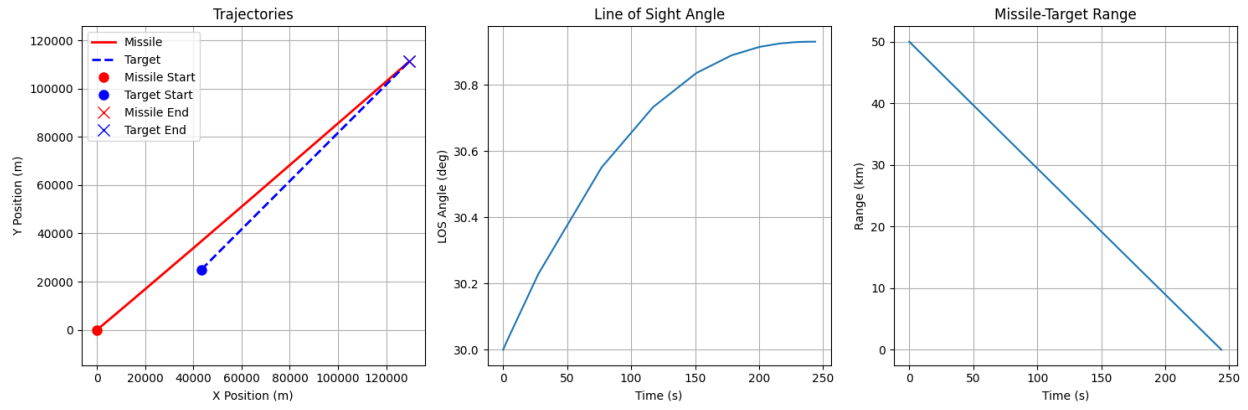
Table 3: Group II: Initial LOS Angle Impact Results

Scenario	θ ($^\circ$)	Intercept Time (s)	Final LOS ($^\circ$)	Efficiency
6	15	229.11	30.93	Best
4	30	243.95	30.93	Good
5	45	252.25	30.93	Good

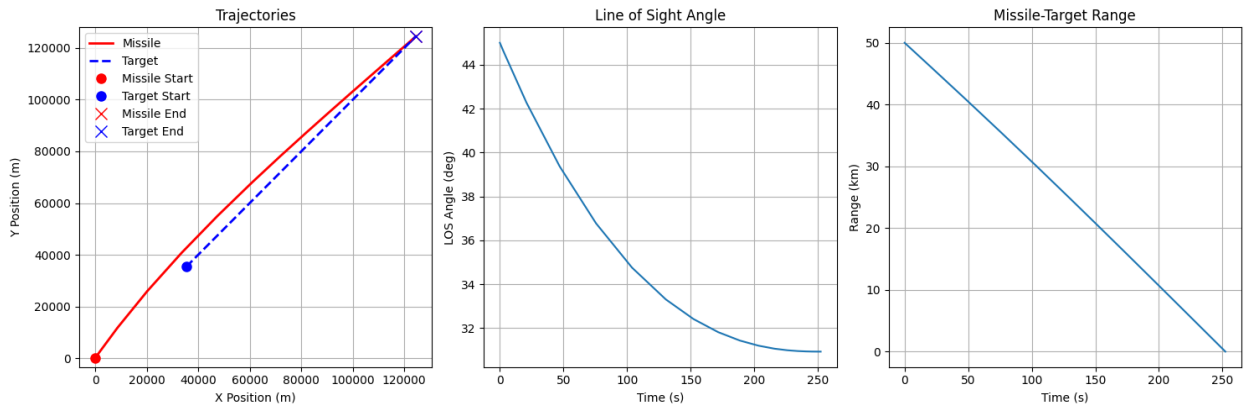
3.2 Key Findings - Group II



(a) $\theta = 15^\circ$ (Scenario 6)



(b) $\theta = 30^\circ$ (Scenario 4)



(c) $\theta = 45^\circ$ (Scenario 5)

Figure 2: Trajectory comparison for different initial LOS angles (Group II)

Initial LOS Angle Effects:

- **Intercept Time:** Significant variation (229-252 seconds, $\sim 10\%$ difference)
- **Final LOS Angle:** Constant at 30.93° regardless of initial LOS

- **Convergence Behavior:** Guidance law drives all scenarios to same final LOS angle
- **Optimal Geometry:** $\theta = 15^\circ$ provides shortest intercept time due to favorable initial geometry
- **Intercept Point:** Varies significantly based on initial geometry

4 Group III Analysis: Target Flight Path Angle Variation (α_T)

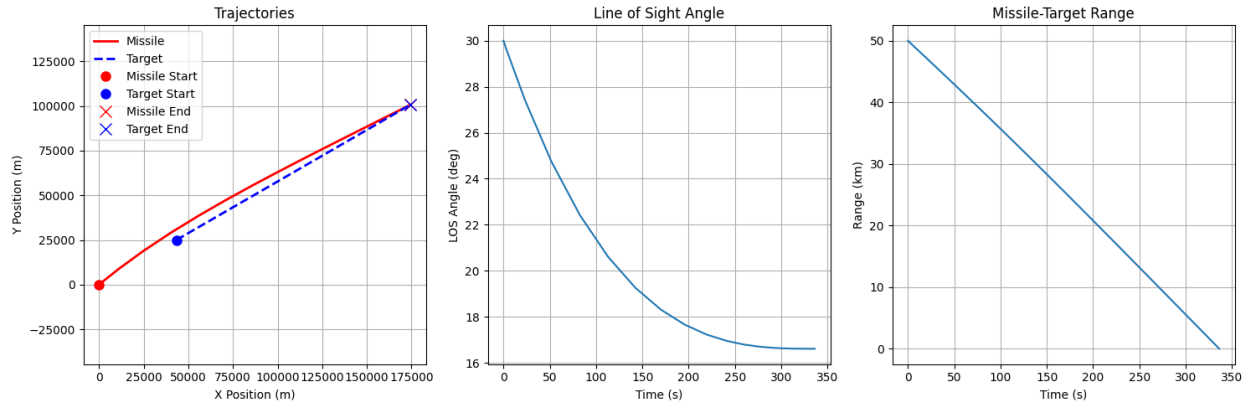
4.1 Experimental Conditions

- Constant parameters: $V_M = 600 \text{ m s}^{-1}$, $V_T = 450 \text{ m s}^{-1}$, $K = 1.333$
- Initial conditions: $\theta = 30^\circ$, $\delta = 10^\circ$
- Variable: $\alpha_T = 30^\circ, 45^\circ, 60^\circ$

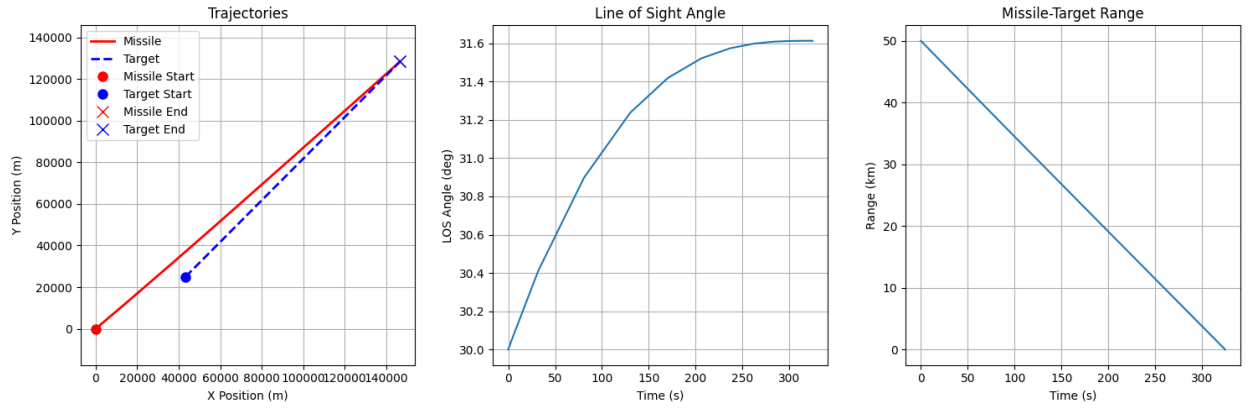
Table 4: Group III: Target Flight Path Angle Impact Results

Scenario	α_T ($^\circ$)	Intercept Time (s)	Final LOS ($^\circ$)	Efficiency
8	60	304.54	46.61	Best
7	45	324.88	31.61	Good
9	30	336.27	16.61	Good

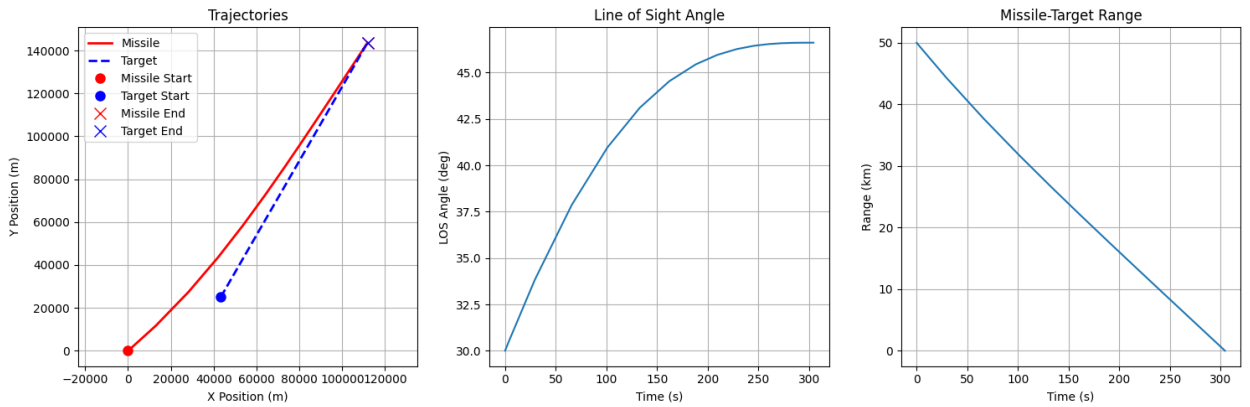
4.2 Key Findings - Group III



(a) $\alpha_T = 30^\circ$ (Scenario 9)



(b) $\alpha_T = 45^\circ$ (Scenario 7)



(c) $\alpha_T = 60^\circ$ (Scenario 8)

Figure 3: Trajectory comparison for different target flight path angles (Group III): (a) $\alpha_T = 30^\circ$ (Scenario 9), (b) $\alpha_T = 45^\circ$ (Scenario 7), (c) $\alpha_T = 60^\circ$ (Scenario 8)

Target Flight Path Angle Effects:

- **Intercept Time:** Significant variation (305-336 seconds, $\sim 10\%$ difference)
- **Final LOS Angle:** Direct correlation with target flight path angle

- $\alpha_T = 30^\circ$: Final LOS = 16.61°
- $\alpha_T = 45^\circ$: Final LOS = 31.61°
- $\alpha_T = 60^\circ$: Final LOS = 46.61°
- **Intercept Geometry:** Higher α_T leads to more vertical intercept points
- **Optimal Condition:** $\alpha_T = 60^\circ$ provides shortest intercept time

5 Group IV Analysis: Low Speed Ratio Boundary Cases

5.1 Critical Performance Limitations

Scenarios 10 and 11 represent the practical operational boundaries of deviated pursuit guidance, demonstrating performance degradation as speed ratio approaches unity.

5.2 Scenario 10: Minimal Speed Advantage

- **Parameters:** $V_M = 500 \text{ m s}^{-1}$, $V_T = 450 \text{ m s}^{-1}$, $K = 1.111$
- **Guidance:** $\delta = 20^\circ$, $\theta = 30^\circ$, $\alpha_T = 45^\circ$
- **Key Result:** Intercept Time = 973.02 s (16.2 min)

Table 5: Scenario 10: Critical Performance Metrics

Metric	Value
Intercept Time	973.02 s
Final LOS Angle	22.66°
Speed Ratio (K)	1.111
Range Reduction Rate	51.4 m s^{-1}
Simulation Points	62
Intercept Position	(352 913 m, 334 612 m)

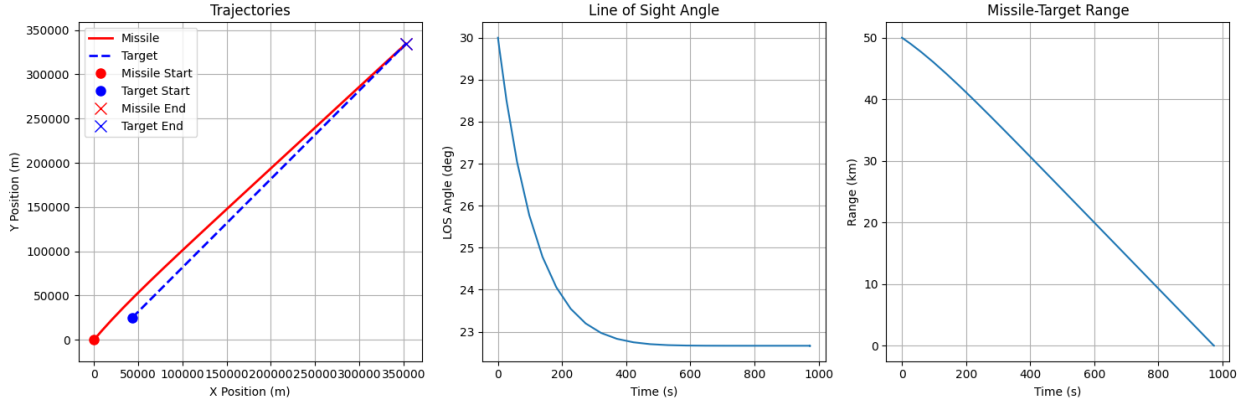
5.3 Scenario 11: Moderate Speed with High Deviation

- **Parameters:** $V_M = 600 \text{ m s}^{-1}$, $V_T = 500 \text{ m s}^{-1}$, $K = 1.200$
- **Guidance:** $\delta = 25^\circ$, $\theta = 30^\circ$, $\alpha_T = 45^\circ$
- **Key Result:** Intercept Time = 493.02 s (8.2 min)

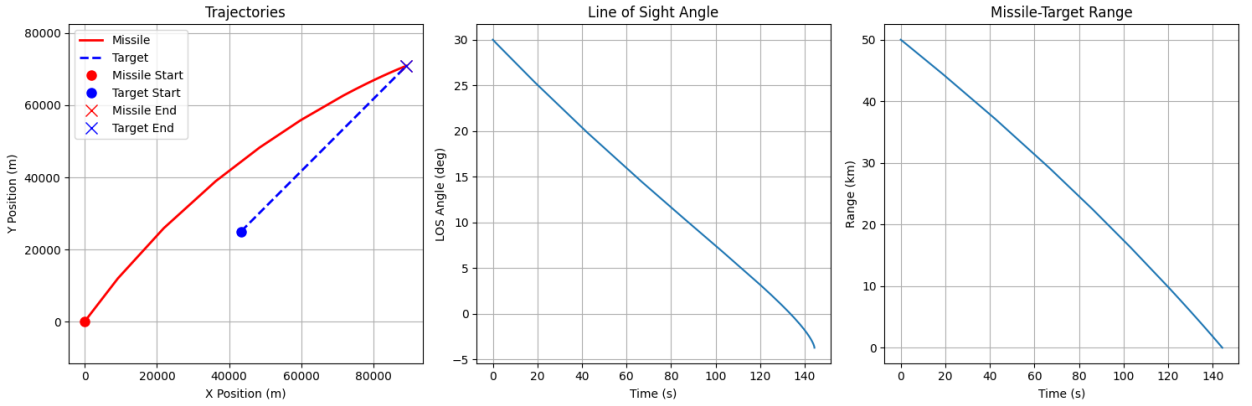
Table 6: Scenario 11: Performance Metrics

Metric	Value
Intercept Time	493.02 s
Final LOS Angle	14.53°
Speed Ratio (K)	1.200
Range Reduction Rate	101.4 m s^{-1}
Simulation Points	48
Intercept Position	(217 610 m, 199 309 m)

5.4 Key Findings - Group IV



(a) Scenario 10: $K = 1.111$, Highly curved trajectory indicating sustained high lateral acceleration



(b) Scenario 11: $K = 1.200$, Moderate curvature with extended intercept time

Figure 4: Low speed ratio trajectory characteristics (Group IV)

Performance Limitations:

- **Exponential Time Increase:** Intercept time increases dramatically as $K \rightarrow 1^+$
- **Trajectory Curvature:** Highly curved paths indicate sustained high lateral acceleration requirements
- **Range Rate Analysis:**
 - Scenario 10: $\dot{R} \approx -51.4 \text{ m s}^{-1}$ (very slow closure)
 - Scenario 11: $\dot{R} \approx -101.4 \text{ m s}^{-1}$ (moderate closure)
- **Practical Implications:**
 - Scenario 10: Likely impractical for real-world applications
 - Scenario 11: Marginal utility, high fuel consumption

6 Comparative Analysis Across Groups

6.1 Parameter Sensitivity Ranking

Table 7: Parameter Sensitivity Analysis

Parameter	Impact on Intercept Time	Impact on Final LOS	Overall Sensitivity
Speed Ratio (K)	Very High	Low	Very High
Target Flight Path (α_T)	Medium	High	High
Initial LOS (θ)	Medium	None	Medium
Deviation Angle (δ)	Low	Very High	Medium

7 Conclusions and Design Guidelines

7.1 Parameter Selection Strategy

1. **Primary Focus:** Maximize speed ratio ($K > 1.3$ for practical implementation)
2. **Deviation Angle Selection:**
 - Use $\delta = 10^\circ - 15^\circ$ for balanced performance
 - Avoid extreme values ($\delta < 5^\circ$ or $\delta > 20^\circ$) to prevent excessive overshoot/undershoot
3. **Engagement Geometry:**
 - Prefer lower initial LOS angles ($\theta < 30^\circ$) when possible
 - Higher target flight path angles ($\alpha_T > 45^\circ$) favor shorter intercept times

7.2 Practical Implementation Recommendations

- **Adaptive Guidance:** Consider variable δ based on engagement phase
- **Acceleration Management:** Monitor lateral acceleration, especially for low speed ratios
- **Mission Planning:** Use pre-engagement analysis to select optimal parameters
- **Performance Boundaries:** Establish operational limits based on comprehensive parameter studies